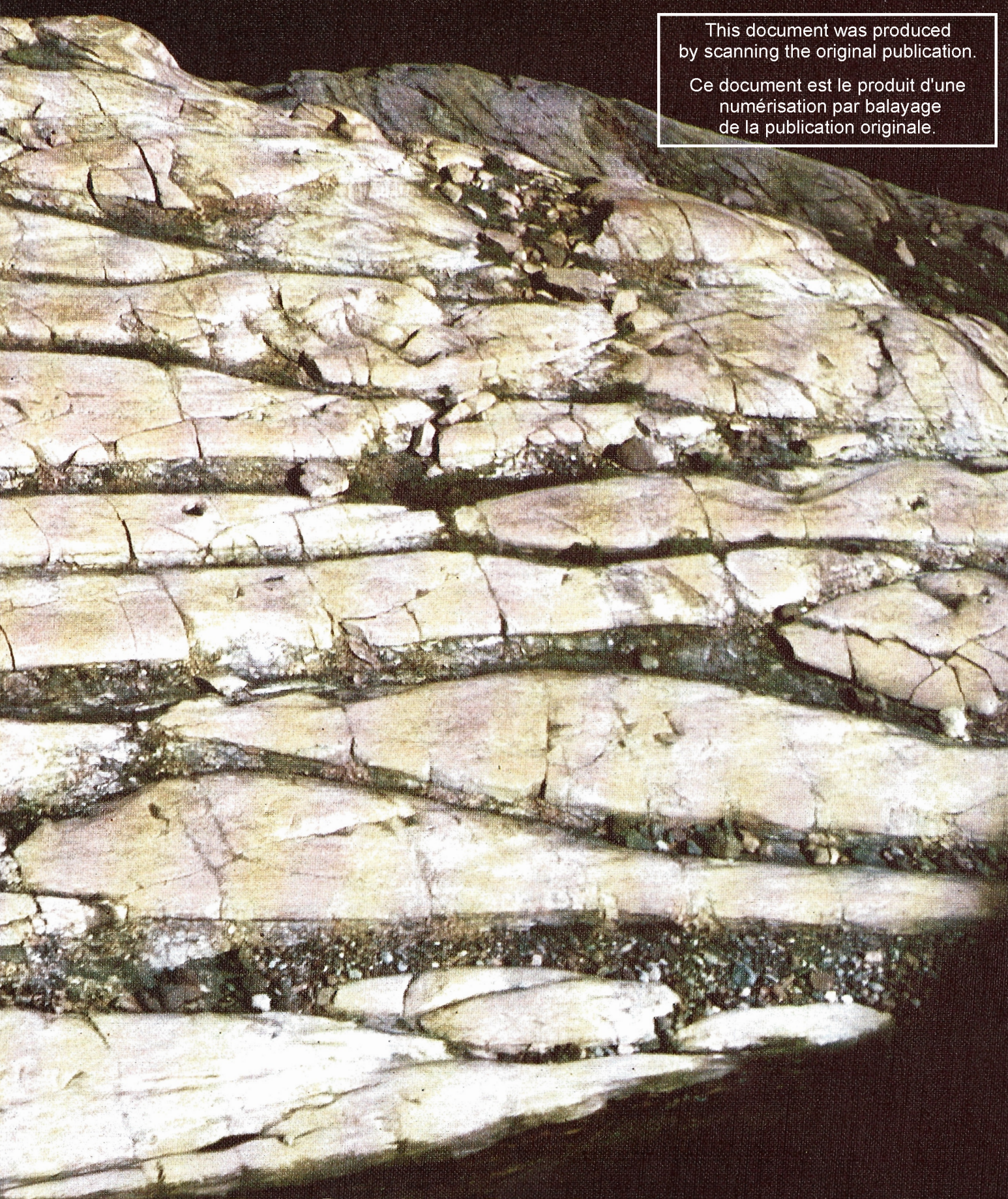
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PROTEROZOIC BASINS OF CANADA

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PROTEROZOIC BASINS OF CANADA

edited by F.H.A. Campbell

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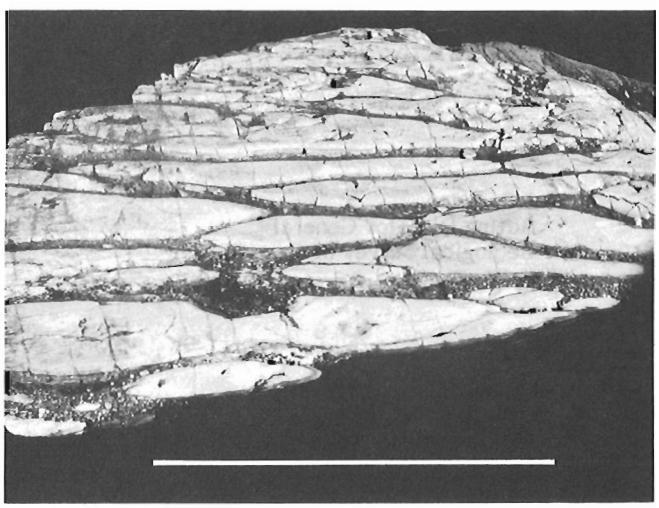
Preface

From time to time the Geological Survey has published the results of symposia of broad, general interest. This allows a wide distribution of the results of these meetings and also makes a much larger audience aware of advances being made in certain fields.

In 1970 we published the proceedings of a symposium on "Basins and Geosynclines of the Canadian Shield". This was probably the first attempt to bring together the results of the reconnaissance studies that had been initiated in the Shield after World War II. In the past decade more detailed mapping projects have been undertaken and our understanding of the Shield has undergone considerable refinement as a result of the application of facies models and plate tectonic theory. It thus seemed appropriate for another symposium to be held, which would allow these advances to be described and discussed. Because much of the work in the past decade has been directed to the study of basinal evolution, particularly in the less highly altered successions, the organizers decided to limit the topic to "Proterozoic Basins of Canada". This highly successful symposium sponsored by the Geological Association of Canada, was held in Halifax in May 1980. Early in the planning for the symposium the Geological Survey was asked to undertake the publication of the proceedings. I am pleased that this was possible and I congratulate the symposium editor and the staff of the Survey's Geological Information Division for producing such an informative and attractive volume.

Ottawa, October 1981

J.G. Fyles
Acting Director General
Geological Survey of Canada



COVER

Aerial view of the mounds of the Reef facies of the Beechey Platform, Western River Formation, Kilohigok Basin. The island outcrop shown in the photograph is in the southeastern part of Bathurst Inlet, N.W.T.

The bar scale represents approximately 100 m.

GSC photograph 203059-H by F.H.A. Campbell.

The funds for the printing of the covers for this volume were generously provided by the Robinson Fund of the Geological Association of Canada.

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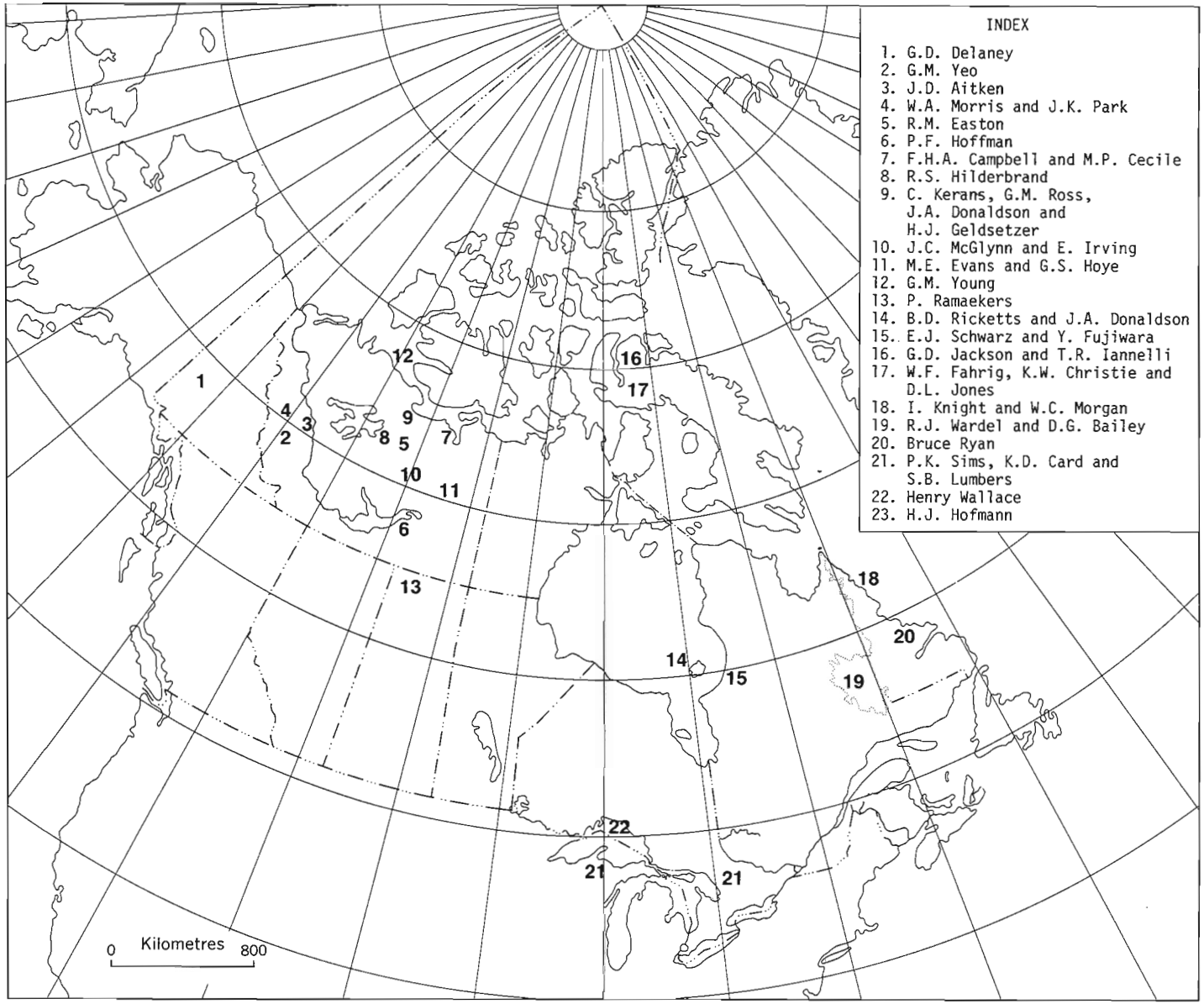
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INTRODUCTION

In 1970 the Geological Survey of Canada published the results of a symposium on "Basins and Geosynclines of the Canadian Shield" (GSC Paper 70-40). The studies on which this symposium was based were largely of a reconnaissance nature but provided the foundation upon which Precambrian geologists have been able to establish local, regional, and continental frameworks. Advances over the past decade have aided in deciphering the basinal evolution of individual successions and relating these to other, typically widely separated sequences. Thus in 1980, a symposium could be devoted solely to Proterozoic sequences in the Canadian Shield and the Cordillera which provided scientists with a forum to discuss and develop their ideas and interpretations. This symposium on "Proterozoic Basins of Canada", was held in Halifax at the Annual Meeting of the Geological Association of Canada and Mineralogical Association of Canada.

The most striking difference between the 1970 and 1980 symposia is the degree to which interpretations of basinal evolution have been affected by the advent of plate tectonics. During the coming decade, as in the past, I am sure the interpretations contained herein will be modified and refined resulting in a comprehensive synthesis of the evolution of Proterozoic Basins in Canada.

The papers in this volume are arranged from west to east (see index map) and in stratigraphic order where papers are from the same region. With the exception of the paper by H.J. Hofmann on Precambrian fossils, the papers cover specific basins and individual authors have gone to considerable lengths to integrate their models into a regional picture. The papers cover specific basins in Canada, the exception being the paper by H.J. Hofmann on Precambrian fossils, and individual authors have gone to considerable lengths to integrate their models into a regional picture.

Acknowledgments

R.T. Haworth, Program Chairman, and organizers of the GAC-MAC Annual Meeting are gratefully acknowledged for their co-operation and assistance in making the symposium a success.

In the two years prior to the meeting Dr. D.J. McLaren, then Director-General of the Geological Survey of Canada, gave his whole-hearted support to the venture.

While I accept full responsibility for the editorial content of the volume, many others aided in reviewing the papers. In alphabetical order these were: J.D. Aitken, W.R.A. Baragar, R.T. Bell, K.D. Card, M.E. Evans, W.F. Fahrig, J.M. Franklin, M.J. Frarey, P.F. Hoffman, M.B. Lambert, J.C. McGlynn, W.C. Morgan, H.C. Palmer, E.J. Schwarz, F.C. Taylor and G.M. Young.

Mrs. B. Cox of the Precambrian Geology Division (GSC) conscientiously typed and retyped versions of the manuscripts. The Photographic and Photomechanical sections of the Geological Survey supplied their customary excellent service in reproducing the illustrations.

Finally, I thank all authors for their perseverance in dealing with a sometimes capricious editor.

En 1970, la Commission géologique du Canada a publié les résultats d'un symposium sur les "Bassins et géosynclinaux du Bouclier canadien" (Étude 70-40 de la C.G.C.). Les études sur lesquelles reposait ce symposium consistaient en grande partie en des travaux de reconnaissance, mais elles ont tout de même permis à des géologues du Précambrien d'établir des structures locales, régionales et continentales. Les progrès accomplis depuis dix ans ont contribué à retracer l'évolution de certaines successions, qui sont ordinairement des séquences largement séparées, et de les mettre en relation les unes avec les autres. Ainsi, en 1980, un symposium a-t-il pu être consacré entièrement aux séquences protérozoiques du Bouclier canadien et de la Cordillère, permettant à des scientifiques de se réunir pour discuter et développer leurs idées et leurs interprétations. Ce symposium, tenu sous le thème de "Bassins protérozoiques du Canada", a eu lieu à Halifax dans le cadre de l'assemblée annuelle de l'Association des géologues du Canada et de l'Association canadienne de minéralogie.

Ce qui frappe le plus dans les différences entre les symposiums de 1970 et de 1980, c'est de voir jusqu'à quel point la tectonique des plaques a modifié les interprétations de l'évolution des bassins. Je suis sûr qu'au cours des dix prochaines années, comme par le passé, nos interprétations d'aujourd'hui seront modifiées et précisées et qu'elles déboucheront sur une synthèse complète de l'évolution des bassins protérozoiques du Canada.

Les études composant ce volume sont groupées géographiquement d'ouest en est (voir la carte-index) et ceux qui s'appliquent à la même région sont classées par ordre stratigraphique. Ces études portent sur des bassins précis du Canada, à l'exception de celle de H.J. Hofmann, qui a pour objet les fossiles du Précambrien; certains auteurs ont dû déployé des efforts considérables pour intégrer leurs modèles dans un tableau régional.

Remerciements

Je remercie chaleureusement R.T. Haworth, président du Programme, ainsi que les organisateurs de l'assemblée annuelle des deux associations pour avoir contribué à faire de ce symposium un succès.

Durant les deux années qui ont précédé la réunion, M. D.J. McLaren, alors directeur général de la Commission géologique du Canada, a soutenu l'entreprise du fond du coeur, et je lui en suis reconnaissant.

Bien que j'assume l'entière responsabilité de la révision du volume, beaucoup d'autres personnes ont contribué à revoir les documents. Ce sont, par ordre alphabétique, J.D. Aitken, W.R.A. Baragar, R.T. Bell, K.D. Card, M.E. Evans, W.F. Fahrig, J.M. Franklin, M.J. Frarey, P.F. Hoffman, M.B. Lambert, J.C. McGlynn, W.C. Morgan, H.C. Palmer, E.J. Schwarz, F.C. Taylor, et G.M. Young.

Je remercie également Mme B. Cox, de la Division de la géologie du Précambrien (C.G.C.), qui a consciencieusement tapé et retapé les diverses versions des manuscrits. J'adresse aussi mes voeux de remerciement aux sections de la photographie et de la photomécanique de la Commission géologique, qui ont reproduit les illustrations avec le brio qu'on leur connaît.

Finalement, je remercie tous les auteurs de la patience dont ils ont fait preuve à l'endroit d'un réviseur parfois capricieux.

F.H.A. Campbell

11.

PALEOMAGNETIC RESULTS FROM THE LOWER PROTEROZOIC ROCKS OF GREAT SLAVE LAKE AND BATHURST INLET AREAS, NORTHWEST TERRITORIES

M.E. Evans and G.S. Hoye
Institute of Earth and Planetary Physics
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Evans, M.E. and Hoye, G.S., Paleomagnetic results from the lower Proterozoic rocks of Great Slave Lake and Bathurst Inlet areas, Northwest Territories; in Proterozoic Basins of Canada, F.H.A. Campbell, editor; Geological Survey of Canada, Paper 81-10, p. 191-202, 1981.

Abstract

Paleomagnetic results are reported from some 300 redbed horizons in the Athapuscow aulacogen and Kilohigok Basin involving the Sosan, Kahochella, and Christie Bay groups in the former and the Western River, Mara, and Peacock Hills formations in the latter. Magnetizations are commonly multiphased, requiring the recognition and removal of recent and ancient overprints before original primary remanences are revealed. For the most part the poles deduced from these formations form a group off the west coast of South America with a mean at 92°W, 14°S (N=8, K=23, A_{95}=12°); no significant polar wander is detectable within the interval represented. This pole (referred to here as the Lower Coronation Pole) indicates that for much of its sedimentation history the Coronation Geosyncline occupied tropical latitudes (about 10°) which is compatible with the abundance of stromatolites and the occurrence of evaporites. In addition to this Lower Coronation Pole, there is a widespread overprint, thought to be associated with uplift and cooling in the Coronation Geosyncline, corresponding to a pole at 091°W, 21°N (K=14, A_{95}=5°).

The pole obtained from the Western River Formation, which is the oldest unit studied, lies off the west coast of Africa, some 80° from the Lower Coronation Pole. One speculative interpretation of this discrepancy involves relative motions between subplates of the Canadian Shield, but it is more likely that it simply represents a time gap of perhaps as much as 200 Ma.

The results from the Stark and Tochatwi formations are aberrant, possibly because of a local tectonic rotation. This suggestion has been tested by collecting new material from the Stark Formation in an area thought to be undisturbed. Unfortunately, the strata there are brecciated and the specimens yield highly scattered directions which do not resolve the rotation problem.

Since the samples were collected in ordered stratigraphic sequences it has proved possible to delineate geomagnetic polarity zones. The magnetostratigraphy of these ancient sedimentary basins is thus beginning to emerge, enabling certain tentative correlations between the various elements of the Coronation Geosyncline to be made.

Résumé

On présente les résultats de levés paléomagnétiques effectués sur environ 300 horizons à couches rouges (red beds), situés dans l'aulacogène d'Athapuscow et le bassin de Kilohigok, englobant les groupes de Sosan, Kahochella et Christie Bay dans l'aulacogène, et les formations de Western River, Mara et Peacock Hills dans le bassin. Généralement, les magnétisations sont multiphasées et exigent l'identification et l'élimination de surimpressions récentes et anciennes, pour laisser apparaître les rémanences primaires originales. En majorité, les pôles déduits de ces formations forment un groupe au large de la côte ouest de l'Amérique du Sud, de coordonnées moyennes 92°W, 14°S (N=8, K=23, A_{95}=12°), et l'on ne peut déceler aucun mouvement notable des pôles correspondant à l'intervalle de temps représenté. Ce pôle (appelé ici pôle de la partie inférieure de la formation de Coronation), indique que pendant une grande partie de son histoire sédimentaire, le géosynclinal de Coronation a occupé des latitudes tropicales (à environ 10°), ce qui concorde avec l'abondance des stromatolites et la présence d'évaporites. Outre ce pôle, on observe une vaste surimpression, que l'on croit associé aux périodes de soulèvement et de refroidissement subies par le géosynclinal de Coronation, et qui est représentée par un pôle de coordonnées 091°W, 21° (K=14, A_{95}=5).

Le pôle obtenu pour la formation de Western River, la plus ancienne unité étudiée, se situe au large de la côte ouest de l'Afrique, à environ 80° du pôle de la partie inférieure de la formation de Coronation. Pour tenter d'expliquer cette discordance, on a considéré les mouvements relatifs entre les sous-plaques du Bouclier canadien, mais il est plus vraisemblable qu'elle correspond à une lacune stratigraphique de peut-être 200 Ma.

Les résultats obtenus sur les formations de Stark et de Tochatwi sont aberrants, peut-être en raison d'une rotation tectonique locale. On a mis à l'épreuve cette suggestion, en recueillant de nouveaux détails sur la formation de Stark, dans une zone que l'on jugeait non perturbée. Malheureusement, les strates de cette formation contiennent des brèches, et les échantillons examinés montrent une importante dispersion de leurs directions, ce qui ne permet pas de résoudre le problème de rotation.

Étant donné que l'on a recueilli les échantillons en suivant le profil stratigraphique, on a pu délimiter les zones de polarité géomagnétique. On commence ainsi à reconnaître la magnétostratigraphie de ces anciens bassins sédimentaires, et donc à établir certaines corrélations entre les divers éléments du géosynclinal de Coronation.

INTRODUCTION

The Athapuscow aulacogen and Kilohigok Basin (Fig. 11.1) both contain several thousand metres of sediments which are for the most part well-exposed and relatively unmetamorphosed. Of particular interest to the present work are the strata of the Great Slave Supergroup of the Athapuscow aulacogen and the Goulbourn Group of the Kilohigok Basin. The formations involved (Table 11.1) are genetically related to the development of the Coronation Geosyncline, which developed along the northwest border of the Slave Province in late Aphebian time, i.e. 2000 to 1750 Ma (Hoffman, 1973; 1980; 1981). Paleomagnetic investigations of several formations have been undertaken to aid in understanding the tectonic evolution of this part of the Canadian Shield and to provide a possible means of correlation between the various elements of the Coronation Geosyncline.

In this paper geological details are kept to a minimum since several excellent accounts already exist (e.g. Hoffman, 1968; Hoffman et al, 1970; Campbell and Cecile, 1976), and others appear in this volume. Procedural details of paleomagnetic field sampling and laboratory work are kept as brief as posible, except where they depart from well-established techniques (see textbooks by Irving (1964) and McElhinny (1973)). The data base for each formation, some of which have already been published, is described briefly before turning to the interpretation and significance of the results.

KILOHIGOK BASIN

New results are reported from three sedimentary formations – the Western River, Mara, and Peacock Hills formations. Data from two sites in a gabbro sill intruding the Western River formation are also described.

Western River Formation

This formation rests unconformably on Archean Yellowknife-like sediments, locally with a regolith and thin conglomerate at the base. These are succeeded by a Lower member, a red siltstone member, a quartzite member, and finally an Upper Argillite member (Campbell and Cecile, 1981). The Red Siltstone member, which ranges in thickness from 100-200 m, was sampled at three sections

Table 11.1. Stratigraphy of the Athapuscow aulacogen and Kilohigok Basin.

		Athapuscow	Kilohigok
GREAT SLAVE SUPERGROUP	Et-Then	Preble Murky	Tinney Cove
	Christie Bay	Unnamed (Hoffman, 1981) Pearson Portage Inlet Tochatwi* Stark*	Amagok Brown Sound Omingmaktook Mbr.
	Pethei	Hearne Wildbread Pekanatui Blanchet McLean Utsingi Taltheilei Douglas Peninsula	Kuuvik
	Kahochella	Charlton Bay* McLeod Bay* Gibraltar* Unnamed (Hoffman, 1981)	Peacock Hills* Quadyuk
	Sosan	Akaitcho River* Kluziai Duhamel Hornby Channel	Mara* Burnside River Western River*
	Seton		

Note: Asterisks indicate formations for which paleomagnetic data are reported in this paper

(Fig. 11.2); section A (140 m; 10 sites = horizons; 34 drill cores), B (53 m; 10 sites; 31 cores), and C (23 m; 5 sites; 15 cores). Thermal demagnetization studies revealed a stable remanence well removed from the present field at the sampling locality at almost all sites (Fig. 11.3, Table 11.2). Most have magnetic vectors pointing eastwards with shallow positive dip, but 5 sites are oppositely polarized. Application of a bedding correction slightly increases the precision parameter (from 11 to 12) but this improvement is not statistically significant, due to the fact that dips are small and fairly constant throughout the region studied.

Mara Formation

This formation generally consists of red siltstones and fine subarkose, and varies in thickness from 100 to 500 m. It was sampled in the uppermost portion immediately underlying the Quadyuk Formation.

The bulk of the samples were collected from a locality on the shores of Bathurst Inlet (107°53'W 67°08'N, Fig. 11.1), but a few samples were also obtained from an area about 100 km to the south (D in Fig. 11.2) that was visited briefly. At this southern locality 5 stratigraphic horizons (15 drill cores) were obtained spread over a thickness of about 500 m. At the Bathurst Inlet locality 43 horizons (155 samples, mostly drill cores) were sampled over about 300 m.

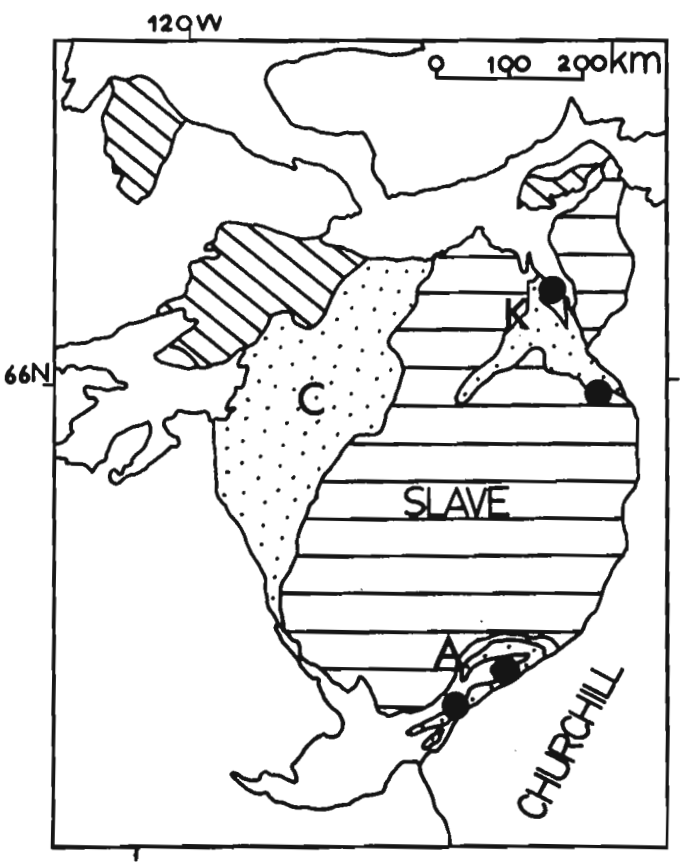


Figure 11.1. Sketch map showing the Slave Province (horizontal lines) and the Coronation Geosyncline with associated aulacogens (dots). Oblique lines represent Helikian and Hadrynian cover rocks. Sampling areas are indicated by large dots.

C = Coronation Geosyncline
K = Kilohigok Basin
A = Athapuscow Aulacogen

Thermal demagnetization was carried out on all samples and most sites yield statistically significant mean magnetic vectors well removed from the present field at the sampling locality (Fig. 11.4, Table 11.3). Two opposed polarity groups emerge but there are also a number of transitional directions, particularly in the upper part of the Bathurst Inlet section, in the 60 m underlying the Quadyuk Formation. These transitional sites are eliminated before averaging the remaining data to obtain a mean direction. In addition rocks at several sites are pale in colour and are very weakly magnetized – these yielded scattered remanence vectors that were highly unstable on demagnetization. It seems that these sites either never had, or have not retained, a remanence measurable with currently available equipment, and they are therefore excluded from further discussion. This leaves 28 site means, of which 9 have mean remanence vectors directed to the south and 19 to the north.

The results from locality D serve to confirm the two polarity groups observed at Bathurst Inlet but the site means are more scattered and the two polarities are not antiparallel. It is possible that some of these sites record

Table 11.2. Site mean results for Western River Formation, after application of bedding correction

Site	N	T	k	D	I	
WAA†	-	-	-	-	-	
WAB	3	600	400	058	+21	
WAC	3	600	22	128	+17	
WAD	3	600	26	079	+15	
WAE	3	600	32	090	+40	
WAF	3	600	220	257	-16	
WAG	3	600	64	111	+27	
WAH	3	600	152	094	+18	
WAI	3	600	52	109	+37	
WAJ	3	600	19	072	+18	
WBA	1	600	-	280	-05	
WBB	2	600	135	266	-25	
WBC	3	600	14	300	-07	
WBD	3	600	6	274	-28	
WBE†	2	660	75	064	+35	
WBF†	-	-	-	-	-	
WBG	3	600	28	076	+24	
WBH	3	600	29	086	+17	
WBI	3	600	168	082	+23	
WBJ	3	600	8	126	+22	
WCA	3	600	113	096	+42	
WCB	3	500	27	065	+39	
WCC	3	600	43	050	+37	
WCD	3	500	9	089	+08	
WCE	3	500	61	040	+30	
MEAN	22 sites		12	086	+25	$\alpha_{95}=9°$

N = no. of cores
T = temperature (°C)
k = Fisher's precision parameter
D = declination (°E of N)
I = inclination (+ downwards)

†These two sites yielded specimen results spread between the two polarity groups indicated by the remaining sites. They appear to contain both polarity vectors that have not been resolved by thermal demagnetization.

N.B. Site WBA is not included in the mean since N is only 1; although its result is in good agreement with the bulk of the other sites.

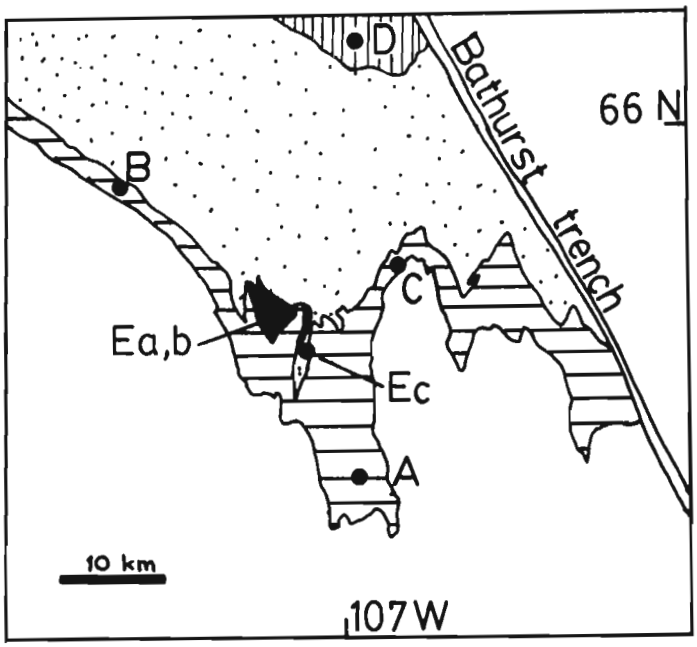


Figure 11.2. Sketch map showing sections sampled in the Western River Formation (A,B,C), gabbro sill (Ea,Eb), and baked contact (Ec).

Horizontal lines = Western River Formation
Dots = Burnside River Formation
Vertical lines = Mara Formation

transitional directions – indeed there is a definite trend in the sequence DA-DB-DC (Table 11.2). With only 5 sites however it is impossible to demonstrate this convincingly, and the D section data are therefore used only as supporting evidence, and are not included in the overall mean for the Mara Formation. This is not a critical decision – inclusion of these 5 sites changes the overall mean given in Table 11.2 by only 3°.

Declination logs (Fig. 11.5) indicate that the lower horizons in both sections are normally polarized (magnetic poles near South America are designated normal for the purposes of this discussion). Between 150 and 200 m below the Quadyuk Formation in the Bathurst Inlet area a reversal takes place, but no sustained northerly group of directions occurs above the transition. Instead there is a complex alternation of normal and reversed horizons. In some cases (e.g. Fig. 11.6) the transition from one polarity state to the other has been recorded. At the D section a reversal also takes place and declination changes from southerly to northerly, but 5 sites are insufficient to reveal finer details. The correlation suggested by Figure 11.5 is possible, but without firmer marker horizons at both localities it must be regarded as promising but tentative.

Peacock Hills Formation

Sixteen sites were samples in a 70 m thick section starting immediately above the Quadyuk Formation in the same locality as the main Mara Formation section. An additional two sites were sampled on a small island 2.5 km to the south. Most of the collection consists of oriented hand samples (48 out of 68) due to difficulties encountered in field drilling. Twelve drill cores were also collected from the Quadyuk Formation, but this stromatolitic carbonate material was very weakly magnetized and yielded unstable, scattered remanence vectors, which are not discussed hereafter.

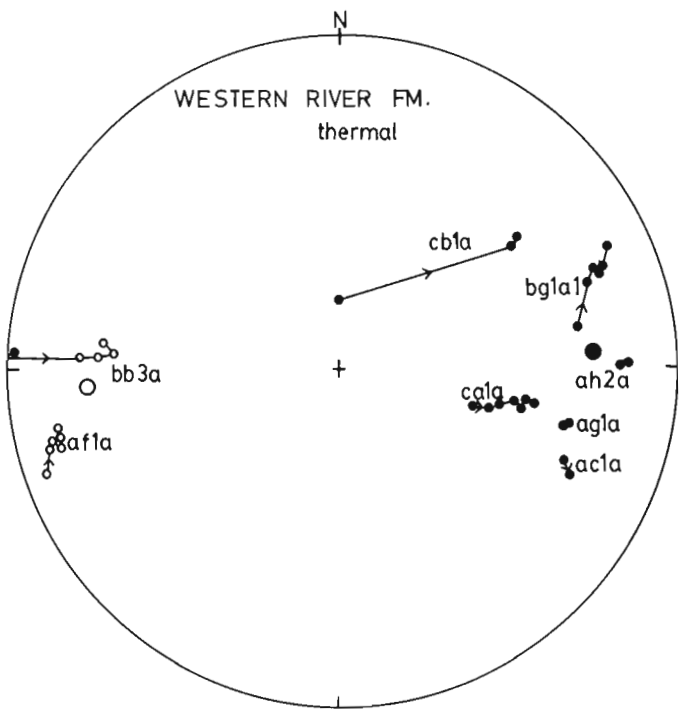


Figure 11.3. Typical thermal demagnetization results for the Western River Formation. Equal area net, solid (open) symbols on the lower (upper) hemisphere. Large symbols represent the overall mean (see Table 11.2).

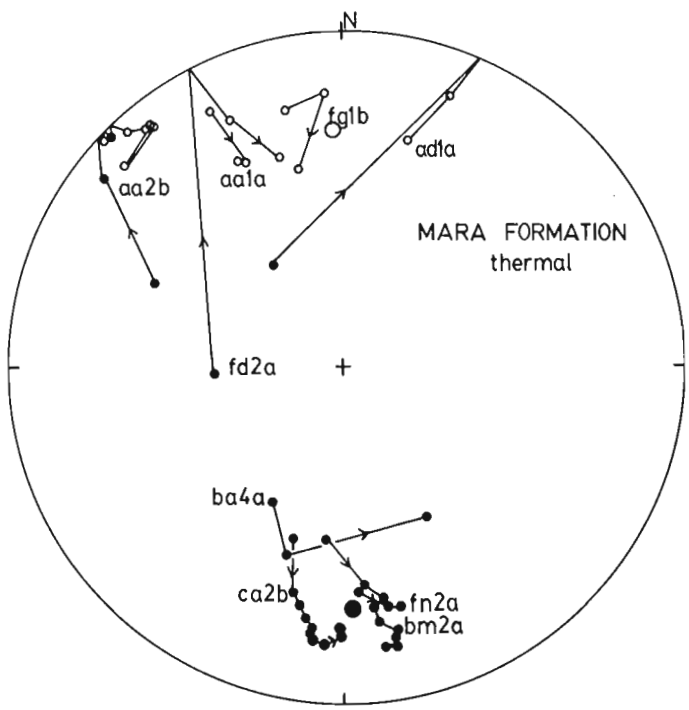


Figure 11.4. Typical thermal demagnetization results for the Mara Formation. Symbols as in Figure 11.3.

During thermal demagnetiztion many Peacock Hills specimens move first to southerly declinations and then abruptly switch to northerly declinations at temperatures between 500 and 700°C (Fig. 11.7); other specimens exhibit northerly directed NRM's which move very little on demagnetization; a few specimens possess stable southerly directions. This type of behaviour suggests the presence of two opposed polarities as described by Roy and Park (1972) and others. It is usually interpreted in terms of a primary remanence with an overprint acquired after one or more field reversals; thus the overprint may not greatly postdate the primary remanence. Roy and Park (1972) have demonstrated that chemical leaching can be an effective technique for resolving the various magnetic components, and we plan to undertake such work with a high-pressure device currently under construction. Thus the Peacock Hills Formation data reported here are of a preliminary nature, with only one sample per site studied in most cases.

Table 11.3. Site mean results for the Mara Formation, after bedding correction

Site	N	T	k	D	I	
BAA	3	500	31	332	-23	
BAC	3	600	188	352	-60	
BAD	4	500	44	018	-03	
BAE	3	600	184	231	+15	
BAF	3	500	11	199	+30	
BAG	3	500	5	163	+01	
BBA	4	500	58	184	+38	
BBB	3	500	46	192	+36	
BBC	3	600	17	164	+39	
BBD	3	500	130	182	+24	
BBE	3	500	32	185	+33	
BBF	3	500	61	183	+32	
BBL	4	500	43	185	+52	
BBM	5	500	46	172	+41	
BBN	4	500	31	179	+55	
BCA	4	500	289	188	+21	
BCB	4	600	153	184	+30	
BCD	4	660	10	003	+15	
8BFC	4	500	9	209	+21	
8BFD	5	600	14	340	-40	
8BFE	4	500	260	166	+28	
8BFF	4	500	122	164	+21	
8BFG	4	600	807	357	-23	
8BFH	4	600	401	338	-09	
8BFI	4	600	12	142	-13	
8BFK	4	600	76	348	-41	
8BFN	4	500	324	176	+26	
8BEA	4	600	867	007	-27	
MEAN	28 sites		12	179	+29	$\alpha_{95}=8°$
DE	3	630	49	325	-05	
DD	3	630	29	338	+17	
DC	3	600	22	146	+38	
DB	3	600	9	182	+55	
DA	3	630	288	154	+71	

- This table excludes 'transitional' sites and sites too weakly magnetized to be reliably measured.
- See Table 11.2 for explanation of symbols.
- Sites starting with D refer to locality D of Figure 11.2.

At this stage we regard the high-temperature results as the best indicator of the "original" magnetization although endpoints are commonly not well-established. The appropriate data are illustrated in Figure 11.8, and summarized in Table 11.4.

Declination logs indicate that the lowermost parts of the sections are normally magnetized, with a reversal taking place within the first few metres above the Quadyuk. With one exception (8BGJ) this polarity is maintained throughout the rest of the section (Fig. 11.9). Further work may remove this exception since the two specimens demagnetized to date agree up to 500°C, (the result given in Table 11.4), but at higher temperatures one swings to a northerly direction as do specimens from the other sites indicating that this site also possesses an underlying reversed magnetization.

Gabbro Sill

Three cores were drilled at each of two sites in a gabbro sill intruding the Western River Formation (Fig. 11.2, Ea, Eb) at a point where these sediments dip at about 5° to the northeast. One site was centrally located and one was near the margin of the mapped outcrop. Site Ec is situated in the Red Siltstone member of the Western River Formation about 20 m from an outcrop of gabbro.

Igneous material was subjected to alternating-field demagnetization and sediments to thermal cleaning. The results are summarized in Table 11.5, and typical demagnetizations are illustrated in Figure 11.10. The remanence vectors move away from the present field towards a shallow northerly direction reminiscent of results from the Mara and Peacock Hills formations (Fig. 11.4, 11.8).

It is important to note that the data from site Ec agree with those of the intrusion but diverge markedly from the other Western River Formation results. This constitutes a convincing baked contact test which strongly favours the view that the magnetization of the intrusion dates from its time of emplacement. Tremblay (1971) reported a K-Ar

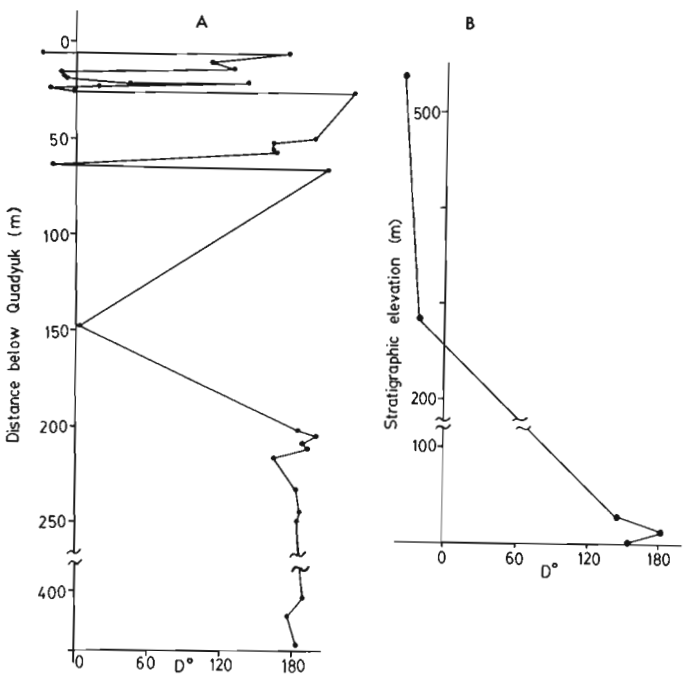


Figure 11.5. Declination magnetograms for the Mara Formation.

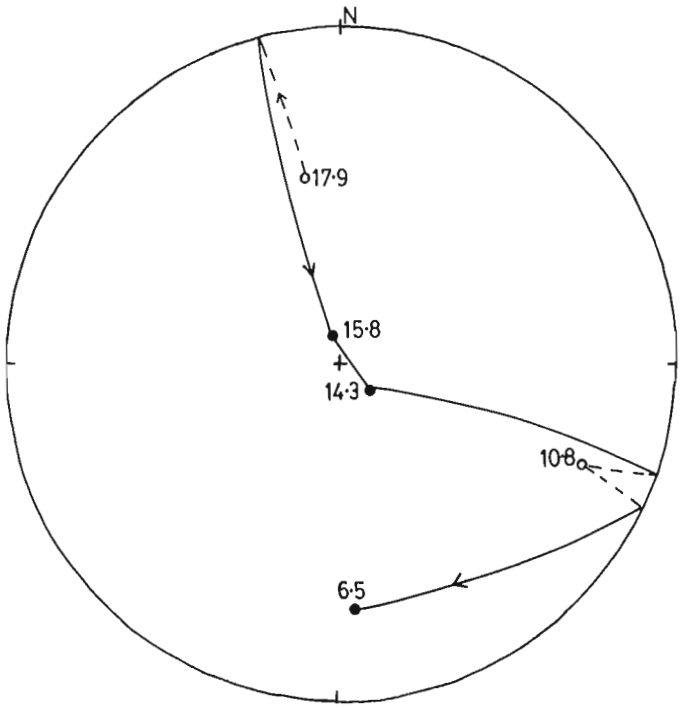


Figure 11.6. The geomagnetic polarity transition recorded between 17.9 m and 6.5 m below the Quadyuk Formation. Symbols as in Figure 11.3.

whole rock age of 1215 Ma for this intrusion which suggests a correlation with the widespread Mackenzie Igneous Episode, but this conflicts with the paleomagnetic evidence. The Mackenzie intrusions have been studied paleomagnetically in areas extending across much of the Shield and yield a tight cluster of poles with a mean at 171°W, 01°N (Irving et al., 1972). The data presented here indicate that the intrusion is magnetized in a direction similar to the Mara and Peacock Hills formations, which yield poles at 107°W, 07°S and 090°W, 15°S respectively (see below). It therefore seems more probable that the intrusion was emplaced about 1800 Ma, and does not, in fact, belong to the Mackenzie Igneous Episode.

ATHAPUSCOW AULACOGEN

Investigations of Athapuscow aulacogen rocks carried out by the University of Alberta paleomagnetic group have appeared in the literature (McMurry et al., 1973; Evans and Bingham, 1976; Bingham and Evans, 1976; Evans et al., 1980; Reid et al., 1981) and discussion is therefore restricted to the points essential for this review.

Akaitcho River Formation

Evans et al. (1980) reported results from 3 sections each about 200 m thick; 40 horizons were sampled, the total number of oriented cores being 120. Thermal demagnetization reveals 2 polarity groups, but these are not exactly antiparallel. This is attributed to a previously recognized widespread magnetic overprint thought to have been acquired during uplift and slow cooling in the Coronation Geosyncline (Irving and McGlynn, 1979). A novel statistical technique was employed to remove this overprint and obtain a best estimate of the underlying original magnetization, which has a mean direction of D=160°, I=+36° (N=35, k=13, α_{95}=7°).

Table 11.4. Site mean results for the Peacock Hills Formation, after dip correction

Site	N	T	k	D	I	
8BGE	1	630	-	298	-10	
8BGG	1	600	-	171	+11	
8BGH	1	500	-	173	+21	
8BGI	1	660	-	342	+02	
8BGJ	2	500	16	188	+25	
8BGK	1	660	-	323	-19	
8BGM	1	660	-	355	+07	
8BGN	1	600	-	324	+09	
8BGO	2	660	12	322	-05	
8BGP	1	660	-	334	+07	
8BGQ	1	600	-	343	+02	
8BGR	1	600	-	353	-11	
8BGS	2	600	22	352	-35	
8BIA	4	600	106	207	+37	
8BIB	4	600	33	342	-27	
MEAN	15		11	342	-12	α_{95}=12°

Symbols are explained in Table 11.2

Kahochella Group

Reid (1972) investigated a thick stratigraphic sequence located on the north coast of Keith Island centred at 111°57'W, 62°03'N. Forty-nine sites were sampled with a typical stratigraphic spacing of 25 m. Some of this work was reported by McMurry et al. (1973) and a full discussion can be found in Reid et al. (1981). After alternating-field and thermal demagnetization a dominant magnetic component in the southeast quadrant with moderate inclinations is observed, with a few sites indicating reversed directions, although never with well-established end-points. As discussed more fully in the next section, Reid (1972) recognized a postfolding overprint which appears to have totally overprinted the original remanence at some sites, but 18 sites retain, at least in part, an earlier remanence from which Reid et al. (1981) deduced a mean direction of D=129°, I=+21° (N=18, k=12, α_{95}=10°). Both polarities occur and are stratigraphically intermingled, with at least 7 reversals implied.

Reid et al. (1981) also reported on five sites in the biogenic, primarily calcareous rocks of the Pethei Group, but poor results from these sites did not provide new information about the ancient geomagnetic field.

Christie Bay Group

Bingham and Evans (1975, 1976) have reported in detail on results from the Stark Formation. They sampled a total of 55 sites, 45 of them located in a single 600 m thick section. The two most significant findings were (a) a profile through an early Precambrian geomagnetic field reversal, and (b) an apparently reliable but highly aberrant paleomagnetic pole. This latter point has been the subject of some debate (Gough et al., 1977; Irving and McGlynn, 1979), the question being whether or not the small area in which the collection was made has undergone rotation about a local vertical axis. There is some structural evidence for this since the bedding strikes in this area diverge 50-60° from the aulacogen as a whole (Bingham and Evans, 1976). Allowance for a rotation of this kind brings the pole into excellent agreement with results from correlative units. Use of such rotations requires extreme caution if circular arguments are to be avoided. We have therefore attempted to resolve this problem by collecting Stark Formation material from another area.

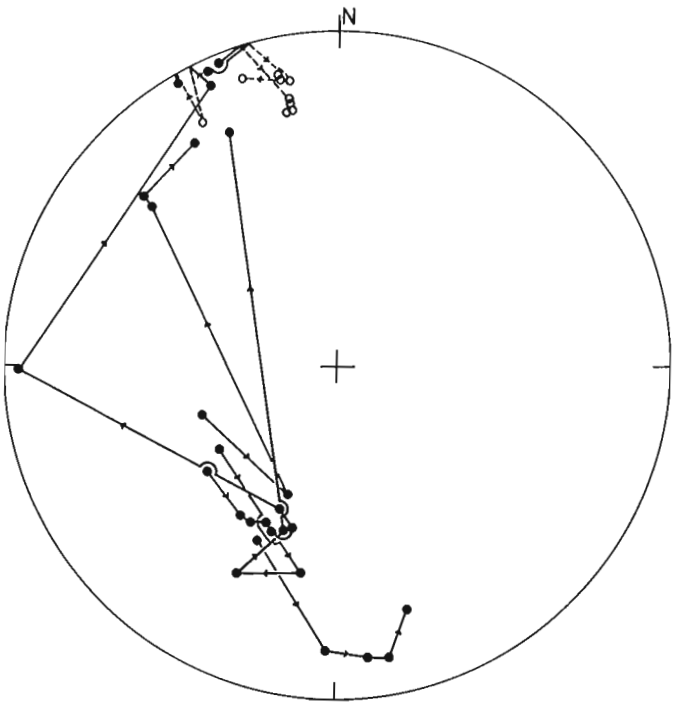


Figure 11.7. Typical thermal demagnetization results for the Peacock Hills Formation. Symbols as in Figure 11.3.

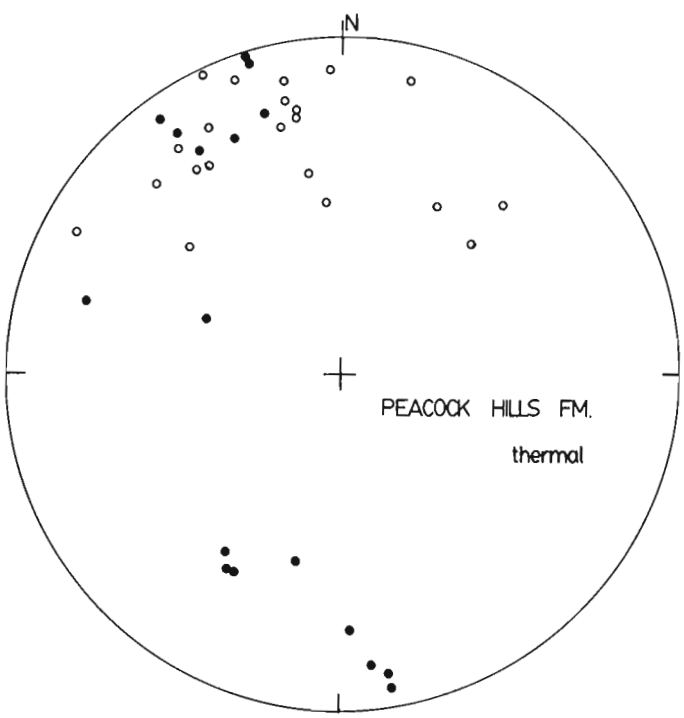


Figure 11.8. Summary of high temperature remanence (600°C and above) in Peacock Hills Formation samples. Symbols as in Figure 11.3.

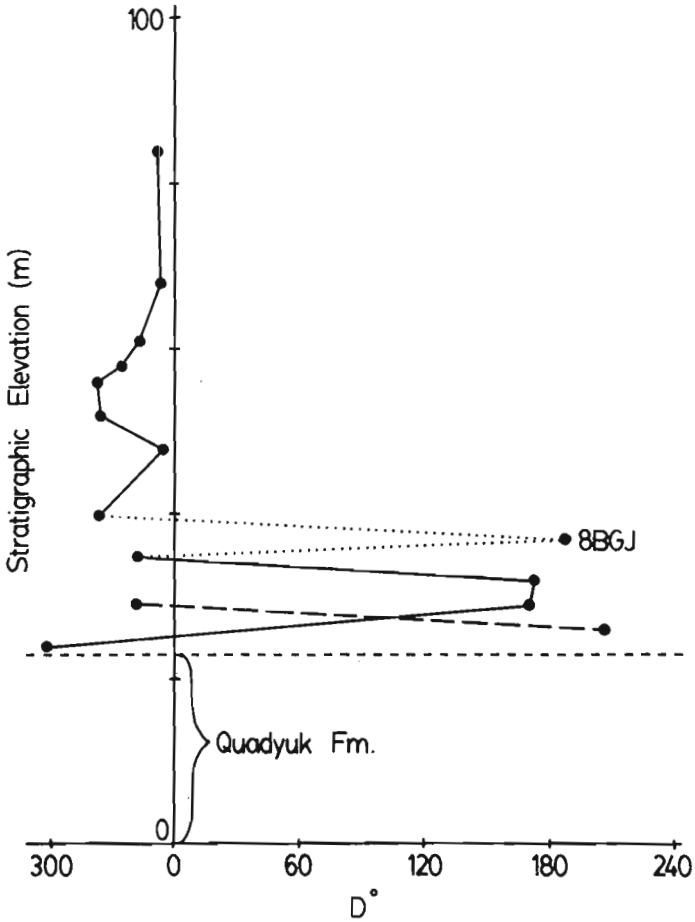


Figure 11.9. Declination magnetograms for the Peacock Hills Formation. The dashed line connects the two sites collected at a separate locality (see text), and the lines to site 8BGJ are dotted as the results from this site are complex (see text).

Eighteen sites were samples on the north shore of Keith Island (111°48'W, 62°06'N) for this purpose. Pilot thermal demagnetization was carried out on 16 specimens, and in most cases acceptable endpoints were reached. The results are very scattered, however, and this remains true when the remaining specimens (after heating to 600°C) are included (Fig. 11.11). This scatter is not only between sites, but also between samples, and even between specimens from a single oriented sample. It almost certainly results from the fact that the Stark Formation here is brecciated, as it is in most outcrops (Hoffman et al, 1977). In fact, one of the few localities where it is not brecciated is the Snowdrift area where the original sampling was done and which yielded the aberrant pole. These new data provide what is essentially a conglomerate test (Graham, 1949), and since the brecciation followed very shortly after deposition (Hoffman et al., 1977) provides tight control on the age of the magnetization of the Stark Formation. However, they obviously do not help to resolve the rotation problem.

Evans and Bingham (1976) have studied the Tochatwi Formation at a total of 29 sites in the Snowdrift area. Most of these possess some kind of magnetic overprint (see below), but 8 sites retain a pre-folding magnetization in close agreement with the Stark Formation and thus yield a similarly aberrant pole.

Table 11.5. Site mean results for the gabbro intrusion and contact sediments

Site	N	T/M	k	D	I
Ea	3	40 mT	46	338	-01
Eb	3	30 mT	226	347	+17
Ec	3	600°C	12	354	+04

Symbols are explained in Table 11.2

M = peak alternating field

Table 11.6. Summary of recent overprints obtained by difference vectors.

Formation	N	R	D	I	k	α_{95}
Tochatwi[1]	16	0-200°C	266	+85	8	14°
Stark[2]	34	0-40mT	347	+87	18	6°
Akaitcho River[3]	33	0-500°C	075	+83	31	5°
Seton[4]	17	0-400°C	073	+86	34	6°

[1]Evans and Bingham, 1976
[2]Bingham and Evans, 1976
[3]Evans et al., 1980
[4]Irving and McGlynn, 1979

N = number of sites
R = temperature or AF range
Other symbols as in Table 11.2

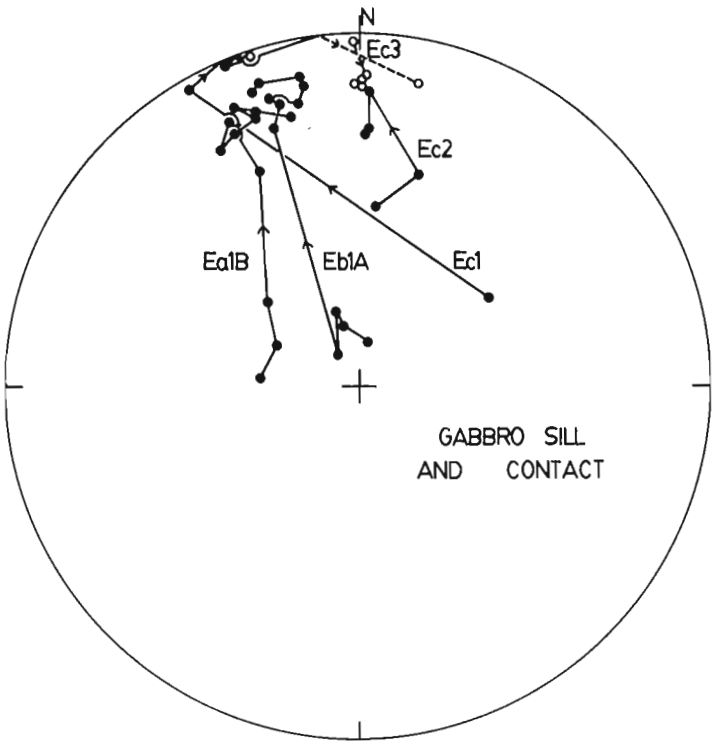


Figure 11.10. Typical thermal (Ec) and magnetic (Ea, Eb) cleaning results from the gabbro sill and baked contact. Symbols as in Figure 11.3.

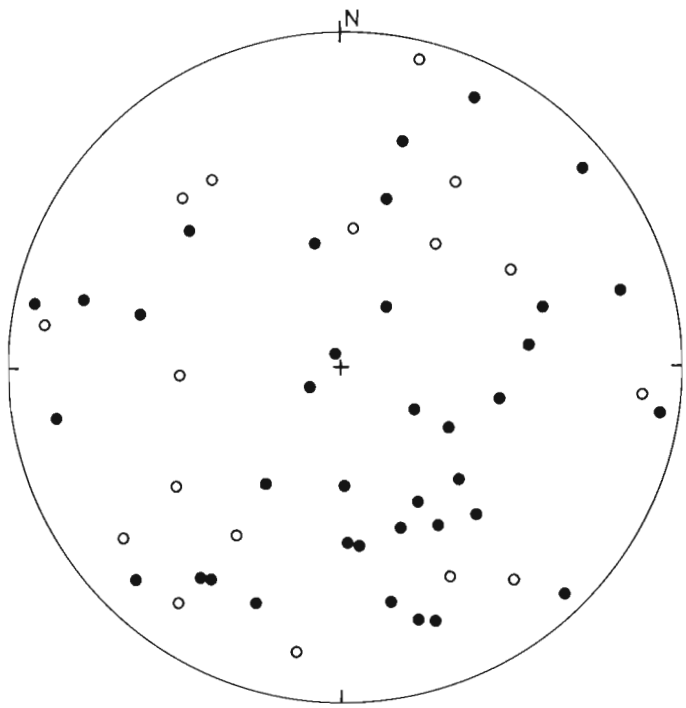


Figure 11.11. Thermal end-point results from brecciated Stark Formation collected on Keith Island (see text). Symbols as in Figure 11.3.

DISCUSSION

Magnetic Overprints

Polyphase magnetization is the rule rather than the exception in these early Proterozoic formations. Detailed demagnetization studies are sometimes effective in isolating the various magnetic components but it is quite probable that many apparently aberrant data are caused by unresolved composite magnetizations.

One effective way of investigating polyphase remanence is to consider the magnetic vectors removed during each step in incremental demagnetization runs. This technique was developed in the early 1970s by E.W. McMurry and A.B. Reid at the University of Alberta and has become a widely accepted procedure in the last decade. It commonly enables one to recognize a recent (probably viscous remanent magnetization, VRM) magnetization acquired in the present ambient field and preferentially carried by grains of low blocking temperature. The Stark, Tochatwi, Akaitcho River, and Seton formations clearly show such an overprint (Fig. 11.12a, Table 11.6). Where geological dips allow a fold test (Stark and Akaitcho River formations) the standard statistical test (Graham, 1949; McElhinny, 1964) demonstrates that this vector was acquired after folding, as expected.

Ancient overprints can also be isolated by vector subtraction, as in the example shown in Figure 11.12b. Here the overprint can be convincingly attributed to heating by a nearby Mackenzie-age intrusion or unexposed equivalent (Evans and Bingham, 1976).

Even where apparently stable remanences occur two (or possibly more) phases may remain. This can sometimes be recognized if reversals occur. A formation which carries a secondary magnetization superimposed on a primary remanence with both normal and reversed polarity generally will yield two polarity groups that are not exactly antiparallel. This is very important to a proper understanding of the Coronation Geosyncline paleomagnetic data. In many of the formations where two polarities are observed these are significantly non-antiparallel. For example the means of the

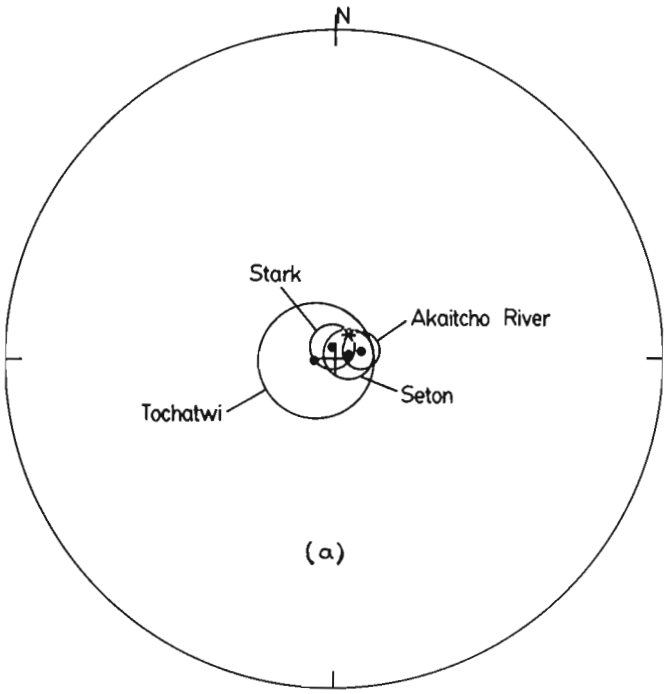


Figure 11.12a. Recent magnetic overprints in Great Slave Supergroup formations. The asterisk represents the present field direction.

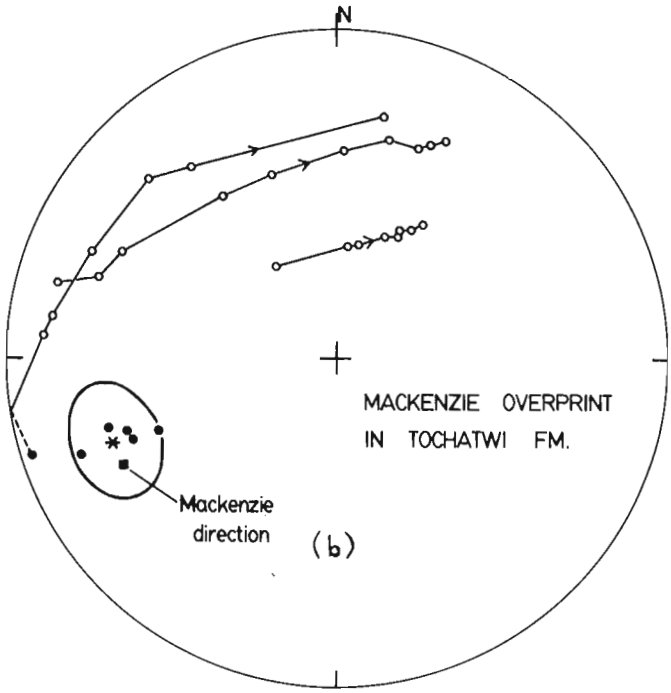


Figure 11.12b. Mackenzie Igneous overprint carried by Tochatwi Formation samples. The curves illustrate typical behaviour during thermal demagnetization. As the Mackenzie overprint is removed the magnetization vectors move gradually towards the primary Tochatwi magnetization (030,-11). The dots represent the magnetic vectors removed between 20 and 500°C during thermal demagnetization, and the asterisk their mean. Symbols as in Figure 11.3.

normal and reversed groups of the Akaitcho River Formation are separated by 142° ± 14° (Evans et al., 1980). Similar results are obtained from the Seton Formation (Irving and McGlynn, 1979) and formations of the Kahochella Group (Reid et al., 1981). The key to this discrepancy is the recognition of an ancient overprint first described by Reid (1972; see also McMurry et al., 1973) and subsequently found by other workers (Evans and Bingham, 1976; Irving and McGlynn, 1979). Difference vectors for moderate to high temperature intervals (mostly between 400° and 600°) form a tight cluster directed steeply downwards in the southeast quadrant. This direction is also recorded by a few sites that seem to have been totally overprinted and can be recognizd without resorting to difference vectors. The relevant data have been compiled by Evans et al. (1980) and their mean lies at D=152°, I=+64° with N=75 sites, k=30, and α_{95}=3°. This overprint is postfolding (Reid, 1972) and is attributed by Irving and McGlynn (1979) to uplift and cooling in the Coronation Geosyncline. Thus we refer to it as the Coronation overprint, although it may, in fact, be much more widspread, extending as far as Greenland (e.g. Morgan, 1976). This overprint was acquired about 1700 Ma, and provides a useful paleomagnetic pole which will be discussed below in the context of apparent polar wander.

Given that the Coronation overprint direction is precisely determined, it is possible to mathematically subtract a suitable vector from those formations exhibiting non-antiparallel reversals to reveal the underlying original magnetization. The details of the methods used are described in Reid et al. (1981) and Evans et al. (1980). The former authors deduced that the thermally cleaned Kahochella samples carry an overprint whose intensity is, on average, 30 per cent that of the underlying 'primary' remanence: the latter deduced a corresponding value of 45 per cent for the Akaitcho River Formation. The 'primary' poles deduced from these 'decontamination' procedures are discussed more fully below.

Magnetostratigraphy

Bingham and Evans (1975, 1976) have reported a detailed study of a geomagnetic polarity reversal in the Stark Formation, and a further example is reported here from the Mara Formation (Fig. 11.6). Such features are of interest to geomagnetists as records of field behaviour from very ancient times. They also offer the possibility of correlation between widely separated, undated, unfossiliferous strata; we have attempted to assess this possibility by collecting from a number of stratigraphically ordered sequences.

Polarity reversals found in the Mara and Peacock Hills formations are summarized in Figures 11.5 and 11.9 in the form of declination magnetograms. They indicate the following simplified sequence – at the base some 200 m of normally magnetized strata (southerly declinations), followed by about 150 m of alternating polarities (at least 9 reversals) up to the Quadyuk Formation, above which the first 60 m of the Peacock Hills Formation is mostly reversely magnetized.

The Akaitcho River Formation of the Great Slave Lake area exhibits a very similar pattern (Evans et al., 1980; Fig. 11.4). It seems likely, therefore, that the reversal which occurs about 200 m below the Quadyuk Formation in the Bathurst Inlet area is equivalent to the one occurring about 100 m above the base of the Akaitcho River Formation in the Great Slave Lake area (Fig. 11.13). This is consistent with the broad correlation suggested by Hoffman et al. (1970), but is by no means compelling. Above this reversal in the Bathurst Inlet area a 150 m gap contains only one sampled horizon. Two possibilities arise from this unfortunate circumstance. One is that the 100 m of mixed polarities plus the 200 m of reversed Akaitcho River Formation strata

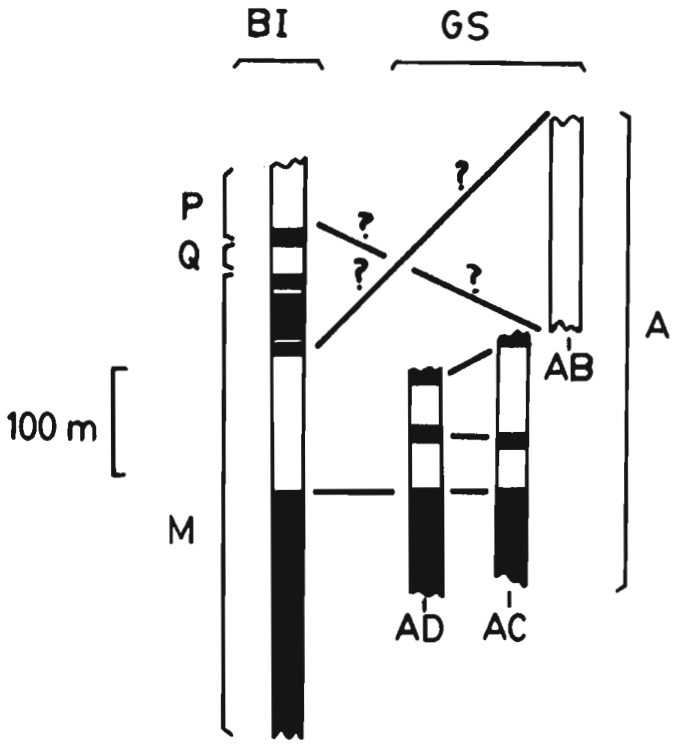


Figure 11.13. Schematic summary of geomagnetic reversals in the Bathurst Inlet (BI) and Great Slave Lake (GS) areas. GS sections are labelled AB, AC, and AD as in Evans et al. (1980)

Black = normal polarity
P = Peacock Hills Formation
Q = Quadyuk Formation
M = Mara Formation
A = Akaitcho River Formation

(section AB, Fig. 11.13) fit into this gap. The other is that the AB section correlates with the Peacock Hills Formation. More detailed sampling is necessary before paleomagnetic data can provide a firm choice between these alternatives.

Paleomagnetic Poles

Poles deduced from the studies reported here are plotted in Figure 11.14 with those from related formations (see Irving and McGlynn, 1979). Most of them form a group with overlapping 95 per cent uncertainty limits, which supports the general synchroneity between developments in the various elements of the Coronation Geosyncline and the tectono-stratigraphic model of Hoffman et al. (1970). Irving and McGlynn (1979) have summarized apparent polar wander (APW) for North America for the interval 2200 to 1700 Ma, using many of these poles. They argue for an APW curve starting in Alaska, sweeping down the west coast, turning east across Chile, then north to Brazil, and finally west to cross itself in Central America. This latter part they refer to as the Coronation Loop. It represents the best summary currently available, although the loop is not as pronounced as they suggest due to minor plotting errors in their Figure 11.

The results from the Great Slave Supergroup, Goulbourn Group, and Epworth Group provide no compelling evidence for coherent polar movement and we therefore represent them collectively as a single "lower Coronation pole" (Fig. 11.15; 092°W, 14°S, N=8, K=23, A_{95}=12°), composed of

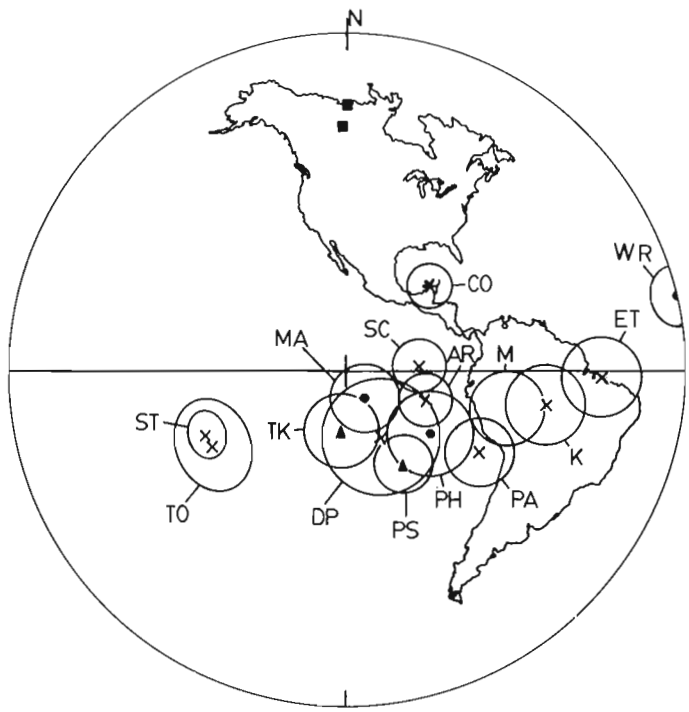


Figure 11.14. Paleomagnetic poles. The asterisk labelled CO represents the Coronation overprint pole (see text). Squares represent the sampling localities.

crosses = Athapuscow Aulacogen
dots = Kilohigok Basin
triangles = Coronation Geosyncline

AR = Akaitcho River Formation
MA = Mara Formation
DP = Douglas Peninsula Formation
ET = Et-Then Group
K = Kahochella Group
M = Martin Formation
PA = Pearson Formation "A"
PH = Peacock Hills Formation
PS = Peninsula Sill
SC = Seton Formation "C" (wrongly labelled SA in Irving and McGlynn, 1979, Fig. 11)
ST = Stark Formation
TK = Takiyuak Formation
TO = Tochatwi Formation
WR = Western River Formation

the data marked with an asterisk in Table 11.7. The Seton "C" magnetization is not included as it consists of two polarity groups which are not antiparallel. Evans et al. (1980) argued that this reflects the presence of a "Coronation overprint" component, which causes a slight northerly shift in the pole. The effect is not large as can be judged from the proximity of the Seton pole to that of the Akaitcho River Formation, which is laterally equivalent (Fig. 11.14; see also discussion above). The result from the Martin Formation of northern Saskatchewan is included in Figure 11.14 to illustrate that this formation is probably correlative to the Great Slave Supergroup, but it is not used in calculating the summary pole shown in Figure 11.15. Poles from the Stark and Tochatwi formations are also excluded because of the still-unresolved structural complication discussed above.

The APW curve can be connected from this lower Coronation pole, through the Et-Then result (Irving et al., 1972), to the Coronation overprint pole also shown in Figure 11.15. The paleo-equator corresponding to the lower Coronation pole is included in Figure 11.15 to emphasize the fact that the Coronation Geosyncline occupied a tropical position when it formed.

The Western River Formation yields a pole which diverges markedly from the other data (Fig. 11.14). At face value it implies intra-cratonic drift, as has been widely suggested for the much younger Grenville Province. It is important to realize that the Western River Formation is older than any paleomagnetically studied unit in the Athapuscow Aulacogen, and that the early part (pre Seton/Akaitcho River times) of the North American APW curve advocated by Irving and McGlynn (1979) includes no data from Coronation Geosyncline rocks. A merging of separate APW curves, and corresponding parts of the Shield, at about 1800 Ma is therefore not out of the question and does not conflict with any currently available data. Much further work, however, is necessary before this can be regarded as anything but highly speculative. An alternative, and perhaps more attractive, interpretation is simply that a significant time gap exists between deposition of the Western River Formation and those formations which yield the Lower Coronation Pole. If typical Phanerozoic rates of polar motion apply, the 77° between the Western River Formation pole and the Lower Coronation Pole could represent some 200 Ma.

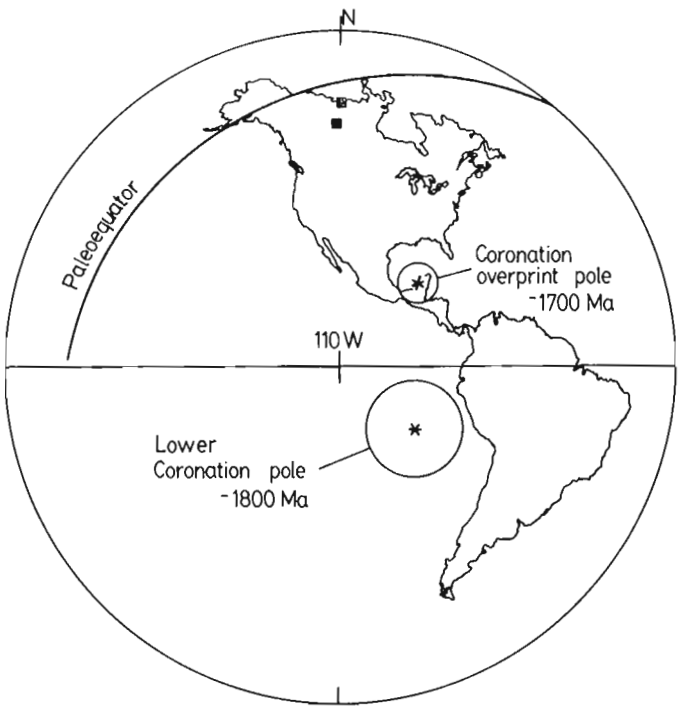


Figure 11.15

The lower Coronation pole and its associated paleoequator, and the Coronation overprint pole.

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Table 11.7. Summary of Coronation Geosyncline and related paleomagnetic results

Formation	N	Directions D,I	k	α_{95}	Poles Long,Lat	K	A_{95}
*Akaitcho River	35	160,+36	13	7°	092W,04S	13	7°
Seton "C"	19	159,+46	24	7°	093W,02N	29	6°
*Kahochella "K"	18	129,+21	12	10°	062W,07S	15	9°
*Douglas Peninsula	6	172,+18	14	19°	102W,17S	19	16°
Stark	39	214,+15	13	7°	148W,15S	9	8°
Tochatwi	8	030,-11	14	15°	143W,18S	24	12°
*Pearson "A"	12	329,-08	21	10°	077W,19S	26	9°
Et-Then	14	294,-21	17	10°	048W,01S	26	8°
Western River	22	086,+25	12	9°	019W,14N	12	9°
*Mara	28	179,+29	12	8°	107W,07S	16	7°
*Peacock Hills	15	342,-12	11	12°	090W,15S	13	11°
*Takiyuak	17	178,+19	15	10°	111W,13S	20	8°
*Peninsula sill	7	345,-02	63	8°	097W,22S	73	7°
Martin	15	333,-29	15	10°	073W,09S	20	9°
Lower Coronation	8	-	-	-	092W,14S	23	12°
Coronation overprint	75	152,+64	30	3°	091W,21N	14	5°

Note: Asterisks indicate formations for which paleomagnetic data are reported in this paper

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